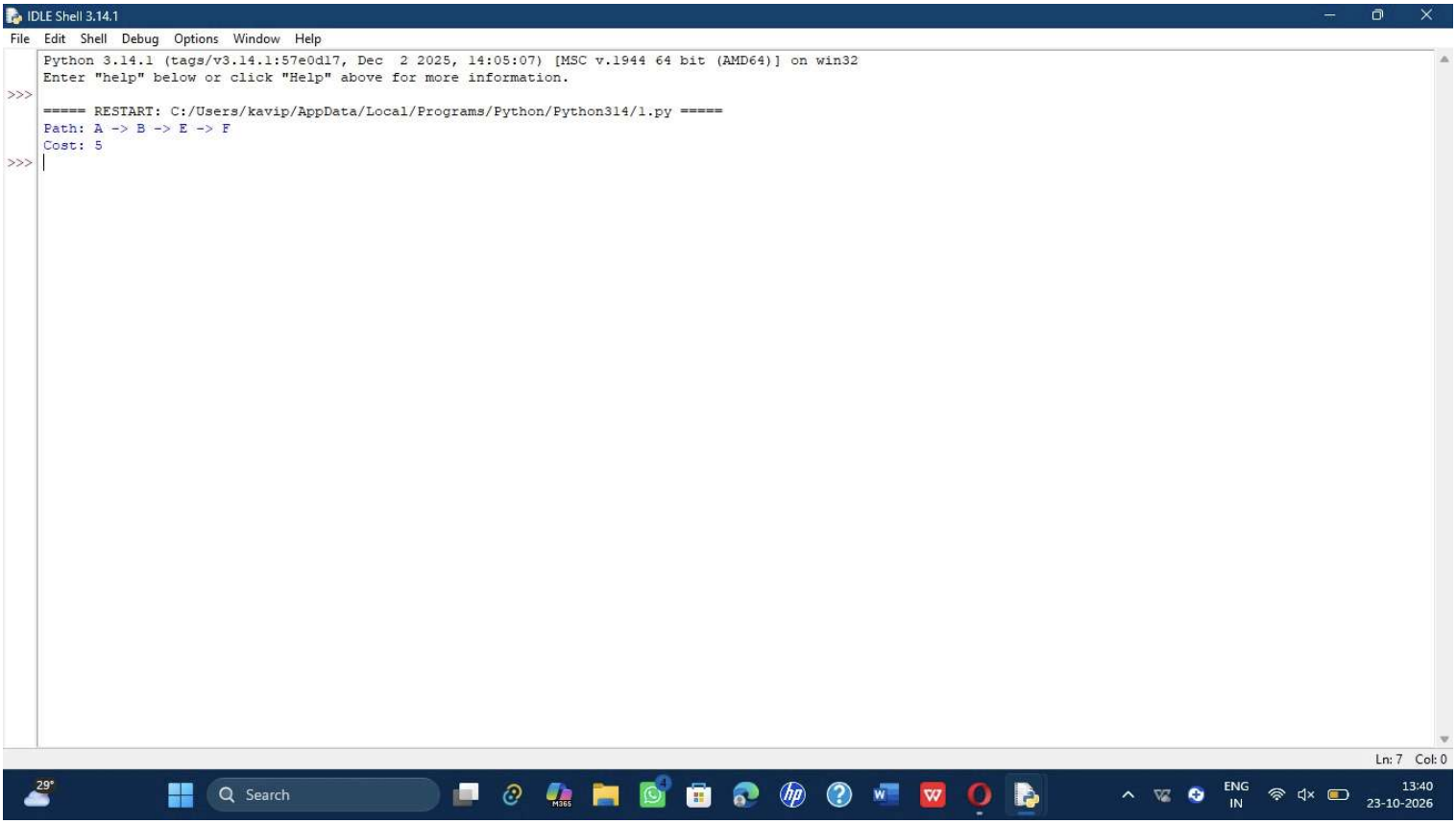

IDLE Shell 3.14.1
File Edit Shell Debug Options Window Help
Python 3.14.1 (tags/v3.14.1:57e0d17, Dec 2 2025, 14:05:07) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/kavip/AppData/Local/Programs/Python/Python314/1.py =====
Path: A -> B -> E -> F
Cost: 5
>>>
Ln: 7 Col: 0
29°
Search
ENG
IN
13:40
23-10-2026

1.py - C:/Users/kavip/AppData/Local/Programs/Python/Python314/1.py (3.14.1)

File Edit Format Run Options Window Help

```
        graph[v] = []

    graph[u].append((v, cost))

start = input("Enter start node: ")
goal = input("Enter goal node: ")

h = {}

for node in graph:
    x, value = input("Enter node and heuristic: ").split()
    h[x] = int(value)

pq = [(h[start], 0, start, [start])]
visited = set()

while pq:
    f, g, node, path = heapq.heappop(pq)

    if node == goal:
        print("Path:", " -> ".join(path))
        print("Cost:", g)
        break

    if node in visited:
        continue

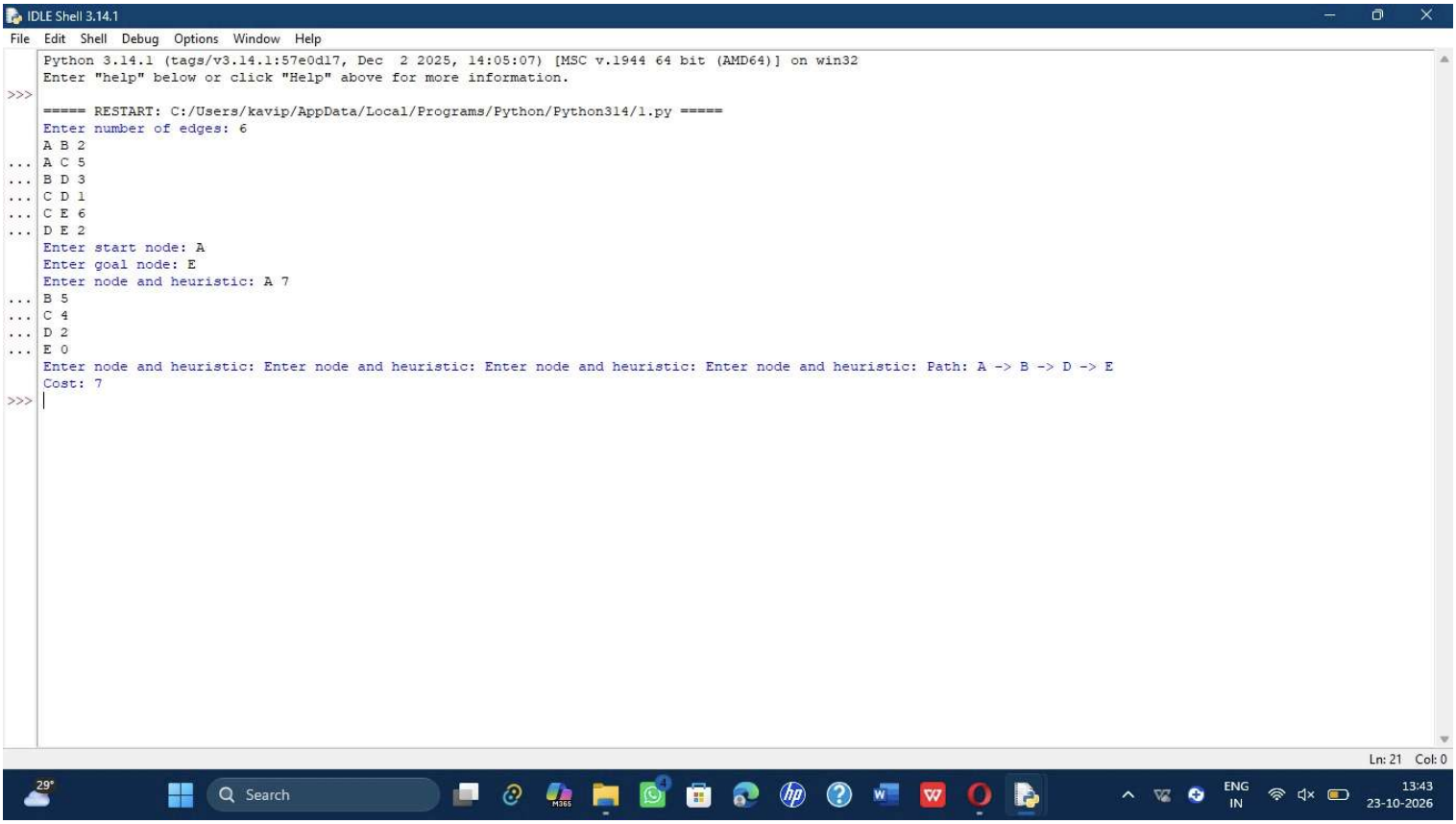
    visited.add(node)

    for neighbour, cost in graph[node]:
        new_g = g + cost
        new_f = new_g + h[neighbour]

        if neighbour not in visited:
            heapq.heappush(
                pq,
                (new_f, new_g, neighbour, path + [neighbour])
            )
else:
    print("No path found")
```

Ln: 54 Col: 0

29° Search ENG IN 13:43 23-10-2026


IDLE Shell 3.14.1
File Edit Shell Debug Options Window Help
Python 3.14.1 (tags/v3.14.1:57e0d17, Dec 2 2025, 14:05:07) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/kavip/AppData/Local/Programs/Python/Python314/1.py =====
Enter number of edges: 6
A B 2
... A C 5
... B D 3
... C D 1
... C E 6
... D E 2
Enter start node: A
Enter goal node: E
Enter node and heuristic: A 7
... B 5
... C 4
... D 2
... E 0
Enter node and heuristic: Enter node and heuristic: Enter node and heuristic: Enter node and heuristic: Path: A -> B -> D -> E
Cost: 7
>>>
Ln: 21 Col: 0
29°
Search
ENG
IN
13:43
23-10-2026

1.py - C:/Users/kavip/AppData/Local/Programs/Python/Python314/1.py (3.14.1)

File Edit Format Run Options Window Help

```
import heapq

n = int(input("Enter number of nodes: "))

graph = []

for i in range(n):
    graph.append(list(map(int, input().split())))

h = list(map(int, input("Enter heuristic values: ").split()))

start = int(input("Enter start node: "))
goal = int(input("Enter goal node: "))

pq = [(h[start], 0, start, [start])]
visited = set()

while pq:
    f, g, node, path = heapq.heappop(pq)

    if node == goal:
        print("Path:", " -> ".join(map(str, path)))
        print("Cost:", g)
        break

    if node in visited:
        continue

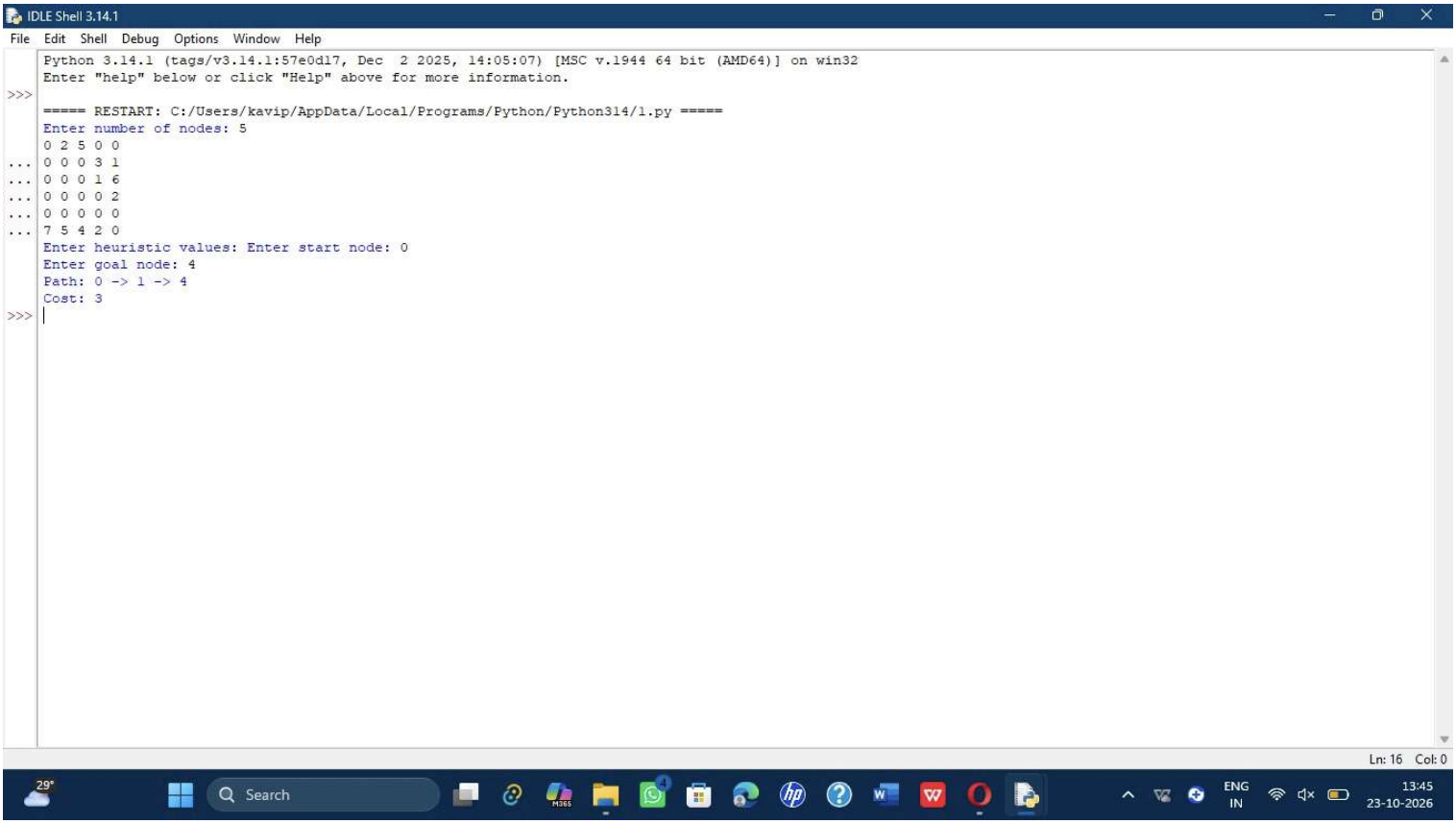
    visited.add(node)

    for neighbour in range(n):
        if graph[node][neighbour] > 0:
            new_g = g + graph[node][neighbour]
            new_f = new_g + h[neighbour]

            heapq.heappush(
                pq,
                (new_f, new_g, neighbour, path + [neighbour])
            )
```

Ln: 1 Col: 0

29° Search ENG IN 13:45 23-10-2026


IDLE Shell 3.14.1
File Edit Shell Debug Options Window Help
Python 3.14.1 (tags/v3.14.1:57e0d17, Dec 2 2025, 14:05:07) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/kavip/AppData/Local/Programs/Python/Python314/1.py =====
Enter number of nodes: 5
0 2 5 0 0
... 0 0 0 3 1
... 0 0 0 1 6
... 0 0 0 0 2
... 0 0 0 0 0
... 7 5 4 2 0
Enter heuristic values: Enter start node: 0
Enter goal node: 4
Path: 0 -> 1 -> 4
Cost: 3
>>>
Ln: 16 Col: 0
29°
Search
ENG
IN
13:45
23-10-2026

1.py - C:/Users/kavip/AppData/Local/Programs/Python/Python314/1.py (3.14.1)

File Edit Format Run Options Window Help

```
    'D': [('F', 4)],
    'E': [('F', 2)],
    'F': []
}

h = {
    'A': 6,
    'B': 4,
    'C': 3,
    'D': 3,
    'E': 1,
    'F': 0
}

start = 'A'
goal = 'F'

pq = [(h[start], 0, start, [start])]
visited = set()

while pq:
    f, g, node, path = heapq.heappop(pq)

    if node == goal:
        print("Path:", " -> ".join(path))
        print("Cost:", g)
        break

    if node in visited:
        continue

    visited.add(node)

    for neighbour, cost in graph[node]:
        new_g = g + cost
        new_f = new_g + h[neighbour]

        heapq.heappush(
            pq,
            (new_f, new_g, neighbour, path + [neighbour])
        )
```

Ln: 17 Col: 0

29° Search ENG IN 13:40 23-10-2026